STUMPRIDER · DESIGN DECISION RECORD

StumpRider TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	SMR-DDR-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stumprider		
Source	docs/decisions/0001-trl2-review-decisions.md @ 6afbf98		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review decisions D1 to D10, decided under Amish's 2026-10-03 pre-approvals

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** decided. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and, for this batch, "Proceed with the remaining 15 scaffolds". Requirements that are not met or at risk are not decided here; they are posed to Amish in `docs/REVIEW.md` and SMR-DEC-001.

> **Safety:** D3, D5 and D9 set safety limits and rules. Each takes the conservative option and names the evidence that would relax it. The tool never replaces a life jacket and is never a reason to keep children on boats.

Context

The scaffold (SMR-PRC-001 v0.1) named seven components and left their form, the pin rating, the canoe it is sized for and the way it is handed out open. These had to be settled to write a measurable precis, a massing model and the calculations.

Options considered and decisions

Table 1. TRL 2 decisions.

#	Question	Options considered	Decision	Why
D1	Form of the rider ring	Spring snap clip; open C ring with a pin across the gap; two hinged halves with a pinned gate	Two half rings of 12 mm steel bar, 100 mm inside, hinged on an M8 bolt and closed by an 8 mm clevis pin with a lanyard	Nothing to corrode shut; opens wide for 4 to 16 mm lines; a smith can make it
D2	Pole material for the prototype	Bamboo; aluminium tube; steel tube	Three 1.45 m sections of 6063-T6 aluminium tube 32 x 2 mm, sleeve joints with lock pins	Uniform stiffness, so trials measure the tool; bamboo stays an option in the R7 and R10 decisions
D3	Shear pin and rating	Plastic, brass or soft aluminium pin; one rating or several	One rating: 2.0 mm soft aluminium wire in double shear, 302 N nominal (256 to 347 N); ten pins from every reel broken on CalRig before use	Highest rating that keeps the design canoe under 5 deg of heel at the top of the band. Conservative; relaxed only by a measured heel test on real canoes
D4	Where the pin sits	Pin for pull only, with a push stop; pin for both	The pin carries both push and pull: the tang stops 45 mm short of the end plate; twisting bears on the fork cheeks	One clear limit in both directions
D5	Operating rules	Free use; rules printed on the tool	Operator kneels; crew slackens the line when the operator lifts; every person in a life jacket; tether end tied to a thwart; cut the net rather than risk a capsize; interim rule: not used from canoes under 7 m or with fewer than three crew until a heel test	Conservative; the interim rule is relaxed only by a heel test with the rated pin
D6	Snags beyond pole reach	Second, longer pole; weight on the cord; weight pinned to the ring	A 1.1 kg weight pinned to the tang in place of the pole head, worked on a 10 m tether	A loose weight would not move the ring
D7	Recovery and marking	Plain cord; float on the cord	6 mm braided polyester cord on a closed-cell foam winder that floats the ring and cord, painted orange	A dropped ring is found and recovered
D8	Net line over the side	Bare gunwale; carved crutch; clamp-on roller	Clamp-on steel saddle with an HDPE roller, for gunwales 30 to 60 mm thick	No drilling of the canoe; no rope cutting into hands or wood
D9	How the tool is handed out	Sale; free handout; with child-protection work	Only through child-protection partners, with the message that nobody dives; first co-design candidates to approach (none approached): Challenging Heights; a Volta landing-site committee through the Fisheries Commission Volta zone; a Ghanaian university fisheries department for trials	The tool must not be read as making child labour safer
D10	Requirements	Change targets; keep	R1 to R10 kept with their targets; design canoe and hard-case dugout defined as assumptions; status given at TRL 3	Targets are Amish's; the calculations report against them

Consequences

- The constructable design (SMR-DDR-002) builds on D1 to D8.
- R1, R3, R7 and R10 come out not met or at risk on paper; they are posed to Amish with options, not decided here.
- Pitch, problem, patent design-arounds and `budget_usd` are unchanged.